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The Black–Scholes Formula

I. THE BLACK–SCHOLES FORMULA

Call price from the FTAP. Given that we know the $\mathbb{Q}$-distribution of S_T, we can write

$$\begin{aligned} c_0 &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{S_T \geq K\}}\big(S_T - K\big)\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{S_T \geq K\}} S_T\right] - e^{-rT} K \mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{S_T \geq K\}}\right] \\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{S_T \geq K\}} S_T\right] - e^{-rT} K \mathbb{Q}[S_T \geq K], \end{aligned}$$

since, in general, for an event A, $\mathbb{E}[\mathbb{1}_A] = \text{prob}[A]$.

Below, we will compute these expectations by brute force. We will refer to $\mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{S_T \geq K\}} S_T\right]$ as "Term 1" and $\mathbb{Q}[S_T \geq K]$ as "Term 2," so

$$c_0 = e^{-rT}\Big(\text{ Term 1 } - K \text{ Term 2 }\Big).$$

Term 2. Since $S_T = S_0 e^{(r-\sigma^2/2)T + \sigma\sqrt{T}z}$, we have that $S_T \geq K$ if and only if

$$z \geq -d_2$$

where

$$d_2 := \frac{\log \frac{S_0}{K} + \left(r - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}.$$

Thus,

$$\text{Term 2} \equiv \mathbb{Q}[S_T \geq K] = 1 - \Phi\big(-d_2\big)$$

where Φ is the standard normal distribution function, i.e. for the standard normal density ϕ,

$$\phi(z) := \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

we have

$$\Phi(\zeta) \equiv \int_{-\infty}^{\zeta} \phi(z)\,dz.$$

Observe that ϕ is an even function (symmetric about zero in the sense that $\phi(z) = \phi(-z)$). This implies that

$$\Phi(-\zeta) = 1 - \Phi(\zeta).$$

You can see this by drawing the density ϕ and noting that the area under the curve below $-\zeta$ coincides with the area under the curve above ζ.[1] So we have

$$\text{Term 2} \;= \Phi\left(d_2\right).$$

[1] One way to see this more formally is by doing a change of variables in the integral with $z' = -z$ so $dz = -dz'$. The key step in the argument is that $\phi(z) = \phi(z')$ because ϕ is even. Thus we have:

$$\begin{aligned}
\Phi(-\zeta) &= \int_{z=-\infty}^{z=-\zeta} \phi(z)\,dz \\
&= \int_{z'=\infty}^{z'=\zeta} \phi(z')\left(-dz'\right) \\
&= \int_{\zeta}^{\infty} \phi(z')\,dz' \\
&= \int_{-\infty}^{\infty} \phi(z')\,dz' - \int_{-\infty}^{\zeta} \phi(z')\,dz' \\
&= 1 - \Phi(\zeta).
\end{aligned}$$

Term 1. We can compute Term 1 as follows:

$$\begin{aligned}
\text{Term 1} &= \mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{S_T \geq K\}} S_T\right] \\
&= \mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\left\{S_0 \exp\left(\left(r-\frac{1}{2}\sigma^2\right)T+\sigma\sqrt{T}z\right)\geq K\right\}} S_0 \exp\left(\left(r-\frac{1}{2}\sigma^2\right)T+\sigma\sqrt{T}z\right)\right] \\
&= S_0 \mathbb{E}^{\mathbb{Q}}\left[\mathbb{1}_{\{z\geq -d_2\}} \exp\left(\left(r-\frac{1}{2}\sigma^2\right)T+\sigma\sqrt{T}z\right)\right] \\
&= \frac{S_0}{\sqrt{2\pi}} \int_{-d_2}^{\infty} e^{\left(r-\frac{1}{2}\sigma^2\right)T+\sigma\sqrt{T}z} e^{-\frac{z^2}{2}}\, dz \\
&= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{-d_2}^{\infty} \exp\left(-\frac{1}{2}\sigma^2 T+\sigma\sqrt{T}z-\frac{z^2}{2}\right) dz \\
&= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{-d_2}^{\infty} \exp\left(-\frac{1}{2}\left[\left(\sigma\sqrt{T}\right)^2-2\sigma\sqrt{T}z+z^2\right]\right) dz \\
&= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{-d_2}^{\infty} \exp\left(-\frac{1}{2}\left[\left(\sigma\sqrt{T}\right)^2-2\sigma\sqrt{T}z+z^2\right]\right) dz \\
&= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{-d_2}^{\infty} e^{-\frac{1}{2}\left(z-\sigma\sqrt{T}\right)^2} dz \\
&= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{z'=-d_2-\sigma\sqrt{T}}^{\infty} e^{-\frac{1}{2}z'^2} dz' \\
&= S_0 e^{rT} \Phi\left(d_2+\sigma\sqrt{T}\right),
\end{aligned}$$

by the same logic as above.

Black–Scholes formula. We can use these calculations to substitute for Term 1 and Term 2 in the expression for the price of the call. We have the following formula:

$$c_0 = S_0 \Phi\left(d_2+\sigma\sqrt{T}\right) - K e^{-rT} \Phi\left(d_2\right)$$

where

$$d_2 \equiv \frac{\log\frac{S_0}{K}+\left(r-\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

We can also express the price of the call at date t by replacing T with the time to maturity $T-t$. This gives the Black–Scholes formula, which is usually written as follows:

$$c_t = S_t \Phi\left(d_1\right) - K e^{-r(T-t)} \Phi\left(d_2\right)$$

where

$$d_1 = \frac{\log\frac{S_t}{K}+\left(r+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

APPENDIX: $\mathbb{P}$- AND $\mathbb{Q}$-DYNAMICS ON THE BINOMIAL TREE

Tree definition. Consider a binomial tree with N steps between date 0 and date T. Call the length of time between each step $h := T/N$. Define the following returns

$$u = e^{\sigma\sqrt{h}} = \frac{1}{d}$$

where the "up" move occurs with probability

$$p = \frac{1}{2} + \frac{m\sqrt{h}}{2\sigma}.$$

Observe that this the two parameters m and σ define the behavior on the tree.

We have the usual no-arbitrage formula for q:

$$q = \frac{e^{rh} - d}{u - d} = \frac{e^{rh} - e^{-\sigma\sqrt{h}}}{e^{\sigma\sqrt{h}} - e^{-\sigma\sqrt{h}}}.$$

Log-nomality. Observe that

$$S_T = S_0 u^k d^{N-k}$$

where k is the number of up moves between date 0 and date T. Thus

$$\begin{aligned}\log\frac{S_T}{S_0} &= \log\left(u^k d^{N-k}\right) \\ &= N\log d + k\log\frac{u}{d},\end{aligned}$$

having used the fact that $\log(ab) = \log a + \log b$ repeatedly. From the definition of the binomial tree, k is binomially distributed (under both $\mathbb{P}$ and $\mathbb{Q}$). By the DeMoivre–Laplace theorem, k is normally distributed for large N. Since $\log(S_T/S_0)$ is of the form $\text{const}_1 + \text{const}_2 k$, $\log(S_T/S_0)$ is normally distributed for large N or, equivalently, for small h.

We now compute the expected return and variance of $\log(S_T/S_0)$ under $\mathbb{P}$ and $\mathbb{Q}$, given the expressions for u, d, and p above.

$\mathbb{P}$-drift. Compute:

$$\begin{aligned}
\mathbb{E}^{\mathbb{P}}\left[\log\left(\frac{S_T}{S_0}\right)\right] &= \mathbb{E}^{\mathbb{P}}\left[N\log d + k\log\left(\frac{u}{d}\right)\right] \\
&= N\log d + pN\log\left(\frac{u}{d}\right) \\
&= N\log e^{-\sigma\sqrt{h}} + pN\log\left(\frac{e^{\sigma\sqrt{h}}}{e^{-\sigma\sqrt{h}}}\right) \\
&= N\left(-\sigma\sqrt{h} + 2\sigma\sqrt{h}p\right) \\
&= \sigma N\sqrt{h}(2p-1) \\
&= \sigma N\sqrt{h}\left[2\left(\frac{1}{2} + \frac{m\sqrt{h}}{2\sigma}\right) - 1\right] \\
&= \sigma N\sqrt{h}\left[\frac{m}{\sigma}\sqrt{h}\right] \\
&= mNh \\
&= mT.
\end{aligned}$$

This is proportional to the time T. The "$\mathbb{P}$-drift" of S is defined as $\mu := m + \frac{1}{2}\sigma^2$, for reasons we'll see below.

$\mathbb{P}$-vol. First note that $p \to 1/2$ as $h \to 0$. Now start with the formula for the variance of a binomial variable and compute:

$$\begin{aligned}
\mathrm{Var}^{\mathbb{P}}\left[\log\left(\frac{S_T}{S_0}\right)\right] &= p(1-p)\left(\log\frac{u}{d}\right)^2 N \\
&= 4p(1-p)\sigma^2 T \\
&\xrightarrow{h\to 0} 4\frac{1}{2}\left(1-\frac{1}{2}\right)\sigma^2 T \\
&= \sigma^2 T.
\end{aligned}$$

Thus, in the limit as there are many trading opportunities (h is small), the $\mathbb{P}$ vol. is σ^2.

$\mathbb{P}$-distribution. In summary, we know the $\mathbb{P}$-distribution of S_T:

$$\log\frac{S_T}{S_0} \overset{\mathbb{P}}{\sim} \mathrm{N}\left[\, mT\,,\, \sigma^2 T\,\right].$$

In other words, variable $\log S_T$ is normally distributed with mean $\log S_0 + mT$ and variance $\sigma^2 T$. This implies that

$$\log S_T = \log S_0 + mT + \sigma\sqrt{T}z, \text{ where } z \overset{\mathbb{P}}{\sim} \mathrm{N}[0\,,\,1].$$

You can check this by observing that $\log S_0 + mT + \sigma\sqrt{T}z$ is normal because z is normal and comput-

ing the mean

$$\begin{aligned}\mathbb{E}^{\mathbb{P}}\left[\log S_0 + mT + \sigma\sqrt{T}z\right] &= \log S_0 + mT + \sigma\sqrt{T}\,\mathbb{E}^{\mathbb{P}}[z] \\ &= \log S_0 + mT + 0 \\ &= \log S_0 + mT\end{aligned}$$

and variance

$$\begin{aligned}\mathrm{Var}^{\mathbb{Q}}\left[\log S_0 + mT + \sigma\sqrt{T}z\right] &= \mathrm{Var}^{\mathbb{Q}}\left[\log S_0 + mT\right] + \left(\sigma\sqrt{T}\right)^2\mathrm{Var}^{\mathbb{Q}}[z] \\ &= 0 + \sigma^2 T \times 1 \\ &= \sigma^2 T.\end{aligned}$$

Now, using the trick that $x = e^{\log x}$, we have that

$$\begin{aligned}S_T &= e^{\log S_T} \\ &= e^{\log S_0 + mT + \sigma\sqrt{T}z} \\ &= S_0 e^{mT + \sigma\sqrt{T}z}.\end{aligned}$$

Hence, using that for x normal, $\mathbb{E}[x] = \exp\left(\mathbb{E}[x] + \frac{1}{2}\mathrm{Var}[x]\right)$, we have

$$\begin{aligned}\mathbb{E}^{\mathbb{P}}\left[\frac{S_T}{S_0}\right] &= e^{mT + \frac{1}{2}\sigma^2 T} \\ &= e^{\mu T},\end{aligned}$$

where μ is the drift defined above.

$\mathbb{Q}$-variance. We can compute the $\mathbb{Q}$-variance as we computed the $\mathbb{P}$-variance above and recover

$$\mathrm{Var}^{\mathbb{Q}}\left[\log\frac{S_T}{S_0}\right] = 4q(1-q)\sigma^2 T$$

recalling the expression for q and noting

$$q = \frac{e^{rh} - e^{-\sigma\sqrt{h}}}{e^{\sigma\sqrt{h}} - e^{-\sigma\sqrt{h}}} \to \frac{1}{2}.$$

as $h \to 0$. You can check this limit with l'Hopital's rule. Take the derivatives of the numerator and denominator with respect to $\sqrt{h}$ and see that

$$\frac{\partial}{\partial\sqrt{h}}\left(e^{r(\sqrt{h})^2} - e^{-\sigma\sqrt{h}}\right) = 2r\sqrt{h}\,e^{rh} + \sigma e^{-\sigma\sqrt{h}} \to \sigma$$

as $h \to 0$ and

$$\frac{\partial}{\partial\sqrt{h}}\left(e^{\sigma\sqrt{h}} - e^{-\sigma\sqrt{h}}\right) = \sigma e^{\sigma\sqrt{h}} + \sigma e^{-\sigma\sqrt{h}} \to 2\sigma$$

as $h \to 0$.

Returning to the expression for the $\mathbb{Q}$-variance, we have

$$\mathrm{Var}^{\mathbb{Q}}\left[\log \frac{S_T}{S_0}\right] = 4q(1-q)\sigma^2 T \to \sigma^2 T.$$

This is the same expression that we found for the $\mathbb{P}$-variance. Thus, *if you can measure the physical variance in the data (i.e. the $\mathbb{P}$-variance), you know the $\mathbb{Q}$-variance that you use to price.*

$\mathbb{Q}$-drift. We can find the $\mathbb{Q}$-drift from the FTAP,

$$S_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T].$$

Again, we use the trick that $x = e^{\log x}$. Here, we can use the formula for the expectation of the exponential of a normally distributed random variable x, $\mathbb{E}[e^x] = e^{\mathbb{E}[x]+\frac{1}{2}\mathrm{Var}[x]}$. We get

$$\begin{aligned} S_0 e^{rT} &= \mathbb{E}^{\mathbb{Q}}\left[e^{\log S_T}\right] \\ &= \exp\left(\mathbb{E}^{\mathbb{Q}}\left[\log S_T\right] + \frac{1}{2}\mathrm{Var}^{\mathbb{Q}}\left[\log S_T\right]\right) \\ &= \exp\left(E^{\mathbb{Q}}\left[\log S_T\right] + \frac{1}{2}\sigma^2 T\right). \end{aligned}$$

Taking logs and rearranging gives

$$\mathbb{E}^{\mathbb{Q}}\left[\log S_T\right] = \log S_0 + rT - \frac{1}{2}\sigma^2 T.$$

$\mathbb{Q}$-distribution. As in the case of the $\mathbb{P}$-distribution, we now can write the $\mathbb{Q}$-distribution as

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,z\right), \text{ where } z \overset{\mathbb{Q}}{\sim} \mathrm{N}[0,1].$$